Lesson notes · for you

Deciding with data

Data science · Level 4 · Lesson 4

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Write a clear data question.
- Design a fair trial.
- Decide with numbers and voices.

Activity: A data story

1. Warm-up (5 min): Name a school rule and say how anyone knows it works.
2. Plan (10 min): With a partner, write a question, the data and a comparison group for one change.
3. Pitch (10 min): Present your plan in one minute; the class asks whose voices are missing.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Why keep a comparison group?
- Whose voices did you include?
- Who makes the final decision?

Check yourself

- Trial: 20 → 14. Comparison: 19 → 17. Work out the effect.

Try it this week
Think of one change you would like at your school, and plan a small fair trial to test it. Present your plan to your class, with the data you would collect.

Worksheet

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Name _____ Class _____ Date _____

1 **The data shows a later start cuts lateness. Who should make the final decision?**

◯ The data ◯ The school's people, using data and voices ◯ The chart

2 **Later-starting schools have better results. Which could be confounders? Choose all that apply.**

☐ The colour of the school gate ☐ Pupils' family income ☐ Urban or rural location

☐ School funding

3 **Compute the difference-in-differences effect and print it.**

```
t_before, t_after = 12, 5
c_before, c_after = 11, 10
effect = t_after - t_before
print(effect)
```

Write your program:

4 **Random assignment balances confounders between groups on average.**

◯ True ◯ False

5 **Why pre-register the question and measures?**

◯ So no consent is needed ◯ To make the trial shorter ◯ To prevent cherry-picking results

✎ **Which job that uses data would you like to try one day? What question would you work on?**

Check your answers

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1 **The school's people, using data and voices**

Data informs; people weigh it and decide.

2 **School funding; Urban or rural location; Pupils' family income**

Each could influence both start time and results.

3

```
t_before, t_after = 12, 5
c_before, c_after = 11, 10
t_change = t_after - t_before
c_change = c_after - c_before
effect = t_change - c_change
print(effect)
```

Lateness fell by 7 in the treated group but would likely have fallen by 1 anyway, so the estimated effect is −6 late arrivals a week, assuming parallel trends.

4 **True**

That is why RCTs give the strongest causal evidence.

5 **To prevent cherry-picking results**

Fixing the analysis plan first keeps the evidence honest.